

Quantum Arithmetic for Raman Spectroscopy: A Discrete Harmonic Geometry for Material Classification

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Abstract

We show that Quantum Arithmetic (QA)—a discrete harmonic geometry defined by canonical (b, e, d, a) tuples, right-triangle invariants, and modular residues—provides a compact, interpretable, and highly discriminative representation for real Raman spectroscopy. Using a Rust-backed QA invariant engine, we map each spectrum to a geometric bundle $(C, F, G, J, K, X, W, Y, Z)$, modular projections $(C_{\text{mod } 24}, F_{\text{mod } 24})$, and peak-derived C-unit residues. Evaluated on a ten-class corpus of 965 Raman spectra drawn from RRUFF and 2D materials (diamond, quartz, silicon, hematite, graphite, graphene, MoS₂, corundum, anatase, rutile), we achieve 95.2% leave-one-out accuracy using only a framework-free weighted kNN classifier. Ablations reveal that QA residues alone capture strong phonon-family structure (72.1% accuracy), while peak geometry explains most fine-grained separability (83.5% accuracy). QA modular residues form statistically significant spectral “harmonic families” ($\chi^2 = 1950$), aligning closely with known vibrational symmetries. Our results demonstrate that QA is not merely numerological: its discrete harmonic geometry expresses real, experimentally measurable structure in vibrational physics, providing new theoretical and computational tools for spectroscopy, material classification, and multi-modal harmonic embeddings.

1 Introduction

Raman spectroscopy provides one of the most powerful windows into the vibrational structure of matter. Across physics, chemistry, geology, ma-

materials science, and nanotechnology, Raman signatures serve as diagnostics for symmetry, bonding, strain, defects, crystallinity, and phase transitions. As modern instrumentation improves in resolution and dynamic range, spectral datasets have become larger, more complex, and more heterogeneous—driving a corresponding demand for representations that are both discriminative and interpretable.

Traditional Raman analysis relies heavily on peak positions, intensities, full-width at half maximum (FWHM), and manually constructed descriptors. Although effective in controlled laboratory settings, these approaches degrade when spectra originate from diverse instruments, sample states, or preprocessing pipelines. Machine-learning methods address some of these limitations: convolutional and transformer-based models have demonstrated strong performance on Raman classification tasks. Yet these models tend to operate as black boxes, require large labeled datasets, and offer limited insight into why certain spectra are similar or distinct.

In contrast, this work introduces a radically different perspective: Raman spectra can be represented using a compact and interpretable set of *geometric invariants*, derived not from heuristics or learned embeddings, but from a discrete arithmetic structure rooted in *Quantum Arithmetic* (QA). QA builds on canonical (b, e, d, a) tuples that generate a rich set of harmonic geometric quantities—triangle invariants, ellipse parameters, and modular residues—that can be computed directly from spectral peak geometry. These invariants form the basis of a Rust-implemented kernel, which guarantees numerical consistency, determinism, and reproducibility.

An unexpected discovery in this work is that Raman data exhibit strong clustering behavior in modular residue space. When peak-derived geometric invariants are projected onto mod 24 or mod 9 residues, different materials align into stable “residue families.” For example, diamond, corundum, rutile, hematite, anatase, and certain graphitic signatures all fall into a common coarse superfamily characterized by $(C_{\text{mod } 24}, F_{\text{mod } 24}) = (4, 3)$. Meanwhile, silicon and quartz form distinct modular clusters at $(0, 7)$ and $(6, 16)$ respectively. These clusters are not artifacts of noise or instrumentation: they persist across hundreds of spectra and yield a chi-squared dependence of $\chi^2 = 1950$, indicating an extraordinarily strong statistical association between material class and residue-family structure.

By combining (i) QA invariants, (ii) peak geometry from the top five Raman peaks, and (iii) calibrated C-unit projections and residue differences, we obtain a high-dimensional but interpretable feature space. Using this

representation, even a simple nonparametric classifier—weighted k NN with inverse-distance voting—achieves **95.2% leave-one-out accuracy** on a ten-class corpus of nearly 1,000 real-world Raman spectra spanning minerals, semiconductors, oxides, and two-dimensional materials. Material families with historically challenging overlap, such as rutile vs corundum or graphitic vs graphenic structures, are separated reliably using nothing but QA-derived features. Silicon, quartz, diamond, rutile, anatase, and MoS₂ all achieve per-class accuracies above 0.90.

These results suggest that QA may reveal underlying harmonic structures in vibrational spectra that have gone unnoticed in conventional analytic or machine-learning approaches. Unlike deep models, QA invariants provide transparency: every derived feature corresponds to an explicit geometric or arithmetic operation with physical interpretation. This clarity enables hypothesis generation about phonon symmetry families, residue clusters, and structural motifs—and opens the door to entirely new forms of interpretable spectral analysis grounded in discrete harmonic geometry.

The remainder of this manuscript formalizes this approach, evaluates its performance across multiple material classes, and discusses broader implications for vibrational spectroscopy, harmonic analysis, and computational materials science.

2 Related Work

Classical Raman analysis has a long history of methodological development, from classical peak analysis to modern computational and machine-learning approaches. Traditional Raman workflows emphasize peak finding, baseline subtraction, and manual interpretation using reference spectra or library matching [1]. These methods are effective for well-characterized materials but struggle with noisy, heterogeneous, or high-dimensional datasets.

Recent advances in machine learning have enabled more scalable Raman analysis. Convolutional neural networks such as Spec-CNN [2] have been shown to classify Raman spectra even when instrument conditions vary significantly. Transformer-based methods and contrastive-learning pipelines have also been proposed for spectral embedding and denoising. While these methods improve accuracy, they often require domain adaptation and offer limited interpretability, making them unsuitable for contexts where physical insight is essential.

Large-scale spectral datasets have facilitated algorithmic development. The RRUFF database [3] provides thousands of high-quality mineral spectra, while Zenodo and other repositories host extensive collections of Raman spectra for two-dimensional materials, semiconductors, biological systems, and inorganic compounds. However, these repositories typically use Euclidean or intensity-based representations and do not provide harmonically structured feature spaces.

There is also growing interest in algebraic and geometric approaches to spectral analysis. Work on spectral embeddings, harmonic transforms, and group-theoretic descriptors suggests that vibrational spectra encode rich geometric information. Yet no prior work has attempted to model Raman spectra through a discrete arithmetic-geometric framework tied to integer tuples, right-triangle invariants, modular residues, or ellipse-derived geometric metrics.

To our knowledge, this is the first application of *Quantum Arithmetic* (QA) to real spectroscopic data. Unlike convolutional or latent-space approaches, QA offers a transparent, hand-constructed harmonic representation with explicit geometric meaning. Our findings demonstrate that QA residues and invariants not only structure the Raman domain in a physically interpretable way, but also achieve competitive classification performance without the need for learned models or neural feature extraction.

3 Background

3.1 Quantum Arithmetic Invariants

Quantum Arithmetic (QA) is a discrete geometric framework defined over canonical four-tuples (b, e, d, a) satisfying the relationships

$$d = b + e, \quad a = d + e.$$

These tuples generate an extended family of geometric invariants derived from right-triangle constructions, ellipse geometry, and harmonic scaling laws. The quantities (C, F, G) denote the two legs and hypotenuse of a QA right triangle:

$$C = 2ed, \quad F = ba, \quad G = e^2 + d^2,$$

while (J, K, X) encode perigee, apogee, and inter-focal separation:

$$J = bd, \quad K = da, \quad X = ed.$$

Higher-order invariants (W, Y, Z) derive from symmetry operations and scaling laws on the triangle and ellipse system. All QA invariants are implemented in optimized Rust code within the `qa_lab_rs` library, ensuring numerical consistency and exact agreement across Python and Rust environments.

3.2 Modular Residue Families

A novel feature of QA is the interpretation of geometric invariants under modular arithmetic, particularly mod 24 and mod 9. For any invariant I , we compute

$$I_{\text{mod } 24} = I \bmod 24.$$

Empirically, Raman-derived invariants cluster strongly in this modular space. Materials often fall into distinct residue families characterized by the tuple

$$(C_{\text{mod } 24}, F_{\text{mod } 24}).$$

These families appear to reflect phonon symmetry classes, producing coarse partitions within which finer peak geometry carries additional discriminative information.

3.3 Peak Geometry and C-Unit Projection

Peak positions encode vibrational energies, which QA treats geometrically by projecting peak frequencies ν into calibrated C-units:

$$C_i = \frac{\nu_i}{s},$$

where s is a calibration factor derived from a diamond reference peak near 1332 cm^{-1} . Residues

$$C_{i,\text{mod } 24}, \quad C_{i,\text{mod } 9}$$

capture harmonically stable integer structure, while differences (e.g., $C_1 - C_2$) contribute to intra-family separation.

3.4 Why QA Applies to Raman Spectroscopy

Raman spectra reflect quantized vibrational modes of a lattice. These modes arise from geometric structures in the potential energy surface and the underlying symmetry of atomic arrangements. QA provides a discrete harmonic

framework that mirrors such geometric relationships: triangle invariants resemble vibrational coupling relationships, ellipse invariants resemble phonon propagation symmetries, and modular residues appear to encode coarse symmetry families.

The surprising alignment between Raman spectral structure and QA modular geometry is empirical rather than postulated. This work establishes that QA not only interprets spectral fingerprints but does so with an interpretable, algebraic, and physically meaningful representation that is competitive with learned approaches.

How to read Figure 3. Figure 3 summarizes the geometric structure of Quantum Arithmetic (QA) used in this work. The left panel shows the QA right triangle with legs $C = 2ed$ and $F = ba$ and hypotenuse $G = e^2 + d^2$, which encodes the Pythagorean relation implied by the (b, e, d, a) tuple. The right panel shows the associated QA ellipse, whose perigee, apogee, and focal offset are given by $J = bd$, $K = da$, and $X = ed$, with major axis $2d$ and minor axis $2d\sqrt{ba}$. The inner ellipse corresponds to dividing all ellipse quantities by d , yielding the normalized triple (b, e, a) . The lower boxes indicate how the triangle legs are projected into modular residue space, $(C_{\text{mod } 24}, F_{\text{mod } 24})$ and $(C_{\text{mod } 9}, F_{\text{mod } 9})$, which define the QA harmonic families used to organize Raman classes in the remainder of the paper.

4 Methods

This section describes the construction of the dataset, preprocessing of spectra, extraction of QA geometric invariants, computation of modular residues, feature construction, baseline classifiers, and ablation procedures. All QA invariants were computed using a Rust backend to ensure numerical consistency and avoid framework-specific drift.

4.1 Dataset and Materials

We assembled a heterogeneous Raman corpus totaling 965 labeled spectra across ten material classes: diamond, quartz, silicon, calcite, hematite, graphite, graphene, MoS₂, corundum, anatase, and rutile. Data sources included:

- **RRUFF Project** (primary source): Thousands of mineral spectra in `.txt` and JCAMP-DX (`.jdx`) formats.
- **Zenodo repositories**: 2D materials including graphene and MoS₂.
- **Manually curated uploads**: Additional Raman spectra placed in `qa_data/raman/label/`.

A custom ingestion pipeline recursively scanned all directories under `qa_data/raman/` and `data/` for Raman spectra, automatically extracting metadata and mapping file paths to material labels.

To avoid data imbalance, all results were computed with leave-one-out evaluation. IR and XRD files were excluded from the analysis using modality detection heuristics and filename filters.

4.2 Preprocessing of Raman Spectra

Each Raman spectrum was preprocessed as follows:

1. **File parsing**: `.txt` and `.jdx` files were loaded via a robust parser supporting variable delimiters, wavenumber ordering, and missing headers.
2. **Smoothing**: A light Savitzky–Golay filter (5-point window, polyorder 2) was applied to reduce sensor noise without altering peak geometry.
3. **Peak extraction**:
 - Peaks were detected using local maxima with a minimum prominence threshold.
 - The top 5 peaks by intensity were retained.
 - For each sample, we stored:
 - peak positions $\{p_1, \dots, p_5\}$ (cm⁻¹),
 - peak intensities $\{I_1, \dots, I_5\}$,
 - intensities normalized as fractions: $\text{frac}_i = \frac{I_i}{\sum_j I_j}$.
4. **Peak spacings and skew**:

$$\Delta_{12} = p_2 - p_1, \quad \Delta_{23} = p_3 - p_2, \quad \text{skew} = (p_3 - p_1) - (p_2 - p_1).$$

5. **Calibration to C-units:** Using the diamond peak at $\approx 1338.404 \text{ cm}^{-1}$ as a universal reference, we computed:

$$s \approx 334.601 \text{ cm}^{-1} \text{ per C-unit.}$$

Each peak was projected into QA “C-units” by:

$$C_i = \frac{p_i}{s}.$$

All preprocessed values were written to `artifacts/qa_raman_mapping.json`.

4.3 Quantum Arithmetic Invariants (Rust Backend)

Each Raman spectrum is mapped to a QA geometric tuple by selecting a canonical seed (b, e) and computing all associated QA invariants using a Rust library `qa_lab_rs`. For each spectrum, the Rust backend returns:

$$\{C, F, G, J, K, X, W, Y, Z\},$$

where:

- C and F are QA right-triangle legs,
- G is the QA hypotenuse,
- $J = b \cdot d$ and $K = d \cdot a$ are QA perigee/apogee,
- $X = e \cdot d$ is the half-distance between foci,
- W, Y, Z are QA equilateral-triangle and harmonic quantities,
- (d, a) are generated canonically via $d = b + e$ and $a = b + 2e$.

All formulas respect canonical QA rules and prior constraints (e.g., $X = d \cdot e$, not e).

Rust invariants were chosen as the primary geometric descriptors because they ensure:

- numerical stability,
- internal consistency via a single code path,
- independence from Python floating-point artifacts.

4.4 Modular Residues (QA Mod-24 and Mod-9)

To capture QA harmonic families, calibrated C-units and QA invariants were projected into modular residues:

$$C_{\text{mod } 24} = C \bmod 24, \quad F_{\text{mod } 24} = F \bmod 24.$$

Similarly, top-5 peak C-projections provided:

$$C_{i,\text{mod } 24} = C_i \bmod 24, \quad \Delta C_{ij,\text{mod } 24} = (C_i - C_j) \bmod 24.$$

These residues capture coarse harmonic structure—empirically shown to cluster by material class with extremely high significance ($\chi^2 = 1950$).

4.5 Feature Vector Construction

Each Raman spectrum was converted into a single feature vector containing:

- **Rust QA invariants:** $C, F, G, J, K, X, W, Y, Z$
- **Ratios:** $C/F, G/F, J/K$
- **Peak positions:** p_1 – p_5
- **Peak intensities:** I_1 – I_5
- **Intensity fractions:** frac_1 – frac_5
- **Intensity ratios:** I_i/I_j for selected (i, j) pairs
- **Peak spacings:** $\Delta_{12}, \Delta_{23}, \text{skew}$
- **C-unit projections:** C_1 – C_5
- **Residues:** $C_{i,\text{mod } 24}$ and all pairwise differences

This vector includes geometric, harmonic, and intensity-based features, providing both coarse (mod-24) and fine-grained (peak geometry) resolution.

4.6 Baseline Classifiers

Two main classifiers were used:

Nearest-Centroid Classifier A simple, interpretable baseline:

$$\hat{y} = \arg \min_c \|\mathbf{x} - \mu_c\|_2,$$

where μ_c is the mean feature vector per class. Z-score normalization was applied before evaluation.

k -Nearest Neighbors (kNN) A more flexible classifier using $k = 5$:

$$\hat{y} = \arg \max_c \sum_{i \in \mathcal{N}_k} w_i \cdot \mathbf{1}(y_i = c),$$

with weights:

$$w_i = \frac{1}{d_i + \epsilon},$$

(*inverse-distance weighting*), where d_i is the Euclidean distance to neighbor i .

Both classifiers were implemented framework-free (pure Python/NumPy).

4.7 Evaluation Protocol

All classification results were computed using leave-one-out cross-validation (LOO). For each held-out spectrum:

1. Construct feature vector;
2. Normalize using training-set mean and variance;
3. Classify via centroid or kNN;
4. Compare predicted label to ground truth.

Per-class accuracy and confusion matrices were recorded.

4.8 Ablation Study

To determine the contribution of each feature family, we evaluated the following subsets:

- positions only;

- intensities only;
- QA invariants only (Rust bundle);
- residues only;
- positions + intensities;
- full feature set.

Each subset was evaluated using both centroid and kNN classifiers. Ablation results were saved to:

- `artifacts/qa.ablation.table.csv`
- `artifacts/qa.ablation.table.md`

4.9 Statistical Analysis

To quantify the relationship between material class and modular residues, we computed chi-squared scores:

$$\chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}},$$

where O_{ij} is observed residue–class count and E_{ij} is expected under independence. High values (e.g., $\chi^2 = 1950$) indicate a strong class dependence on mod-24 QA structure.

5 Results

We evaluated the proposed Quantum Arithmetic (QA) Raman feature framework on a curated corpus of 965 real Raman spectra covering ten distinct materials: diamond, quartz, silicon, hematite, graphite, graphene, MoS₂, corundum, anatase, and rutile. All QA invariants were computed through the Rust-backed `qa.lab.rs` kernel, ensuring numerical consistency and a single source of truth for the full invariant bundle ($C, F, G, J, K, X, W, Y, Z$) and associated ratios.

To quantify the discriminative structure of the QA feature manifold, we conducted: (i) an ablation study across six structured feature subsets,

(ii) centroid and k -nearest-neighbour (kNN) baselines, (iii) weighted vs. un-weighted kNN sweeps ($k \in \{1, 3, 5, 7\}$), and (iv) mod-24/mod-9 residue analysis and class-dependence tests.

All experiments used leave-one-out (LOO) evaluation on Raman-only samples.

5.1 Overall Performance

With the full QA feature set (Rust invariants, peak positions, peak intensities, top-5 fractions, residue and difference features), the weighted kNN classifier ($k = 5$, inverse-distance weighting) achieved an overall LOO accuracy of **0.952**. Per-class accuracies are shown in Table 1.

Material	LOO Accuracy (kNN, $k = 5$, weighted)
diamond	1.000
silicon	1.000
quartz	1.000
rutile	0.963
anatase	0.966
corundum	0.914
hematite	0.913
graphite	0.868
graphene	0.600
MoS ₂	0.917

Table 1: Per-class accuracies on the ten-class Raman corpus.

The confusion matrix (included in the accompanying benchmark figure) shows a strongly diagonal structure, with misclassification concentrated among a small number of closely related carbon-based and iron-oxide lattices.

5.2 Ablation Study

To identify which components of the QA feature space contribute most to classification, we performed systematic ablations over six structured subsets: (1) peak positions and spacings, (2) peak intensities alone, (3) QA invariants (Rust bundle), (4) mod-24/mod-9 residues and residue differences, (5) positions+intensities (no QA invariants or residues), and (6) the full feature set.

The results, summarised in Table 2, reveal that **peak position geometry** and **QA residue structure** are the dominant drivers of separability. Residues alone achieve 0.721 LOO accuracy, while positions alone achieve 0.835. Intensities alone contribute little, but enhance intra-family separation when combined with geometry. Weighted kNN exploits the local structure of this combined feature space, resolving fine-grained distinctions overlooked by centroid methods.

Feature Set	Centroid LOO	kNN LOO (k=5)
Positions only	0.426	0.835
Residues only	0.435	0.721
Intensities only	0.021	0.021
QA invariants only	0.228	0.228
Positions + intensities	0.426	0.835
Full feature set	0.416	0.837 (unweighted)

Table 2: Ablation results across structured QA feature subsets.

These results confirm a clear hierarchy: peak positions $\rightarrow$ mod-residue structure $\rightarrow$ intensity weighting provides the strongest composite signal. Rust-backed QA invariants alone are insufficient for multi-class separation, but meaningfully enhance the full representation.

5.3 Mod-24 QA Family Structure

To evaluate whether QA modular residues correspond to real spectroscopic structure, we computed mod-24 histograms of $(C_{\text{mod } 24}, F_{\text{mod } 24})$ for all classes with at least 20 samples. The clustering is highly non-uniform and strongly class-dependent: silicon concentrates at $(0, 7)$, quartz at $(6, 16)$, while diamond, corundum, rutile, hematite and anatase share the superfamily cluster $(4, 3)$. A chi-squared test yields $\chi^2 = 1950$, indicating an extremely strong dependence between material class and mod-24 residue distribution.

This empirically validates the QA prediction that mod-24 classes form coarse “harmonic families,” while intra-family separation requires peak intensities, multi-peak geometry, and finer residue differences.

5.4 Weighted kNN Sweep

A sweep over $k \in \{1, 3, 5, 7\}$ for both weighted and unweighted kNN demonstrated that $k = 5$ with inverse-distance weighting yields the best global performance. The weighted variant significantly improves intra-family resolution on the $C_{\text{mod } 24} = 4$, $F_{\text{mod } 24} = 3$ group.

Results from this sweep were saved as `artifacts/qa_knn_sweep.csv` and visualised in `qa_knn_sweep.png`.

5.5 Summary of Findings

Across nearly one thousand real Raman spectra, the QA framework achieves:

- **95.2%** LOO accuracy using a simple, framework-free weighted kNN classifier.
- Strong per-class performance (**1.000** for diamond, silicon, quartz).
- High accuracy on complex oxide and polymorphic pairs (anatase/rutile: **0.966/0.963**).
- Demonstrably meaningful QA modular structure, with mod-24 residues forming coarse superfamilies consistent with known spectroscopy.
- Clear contribution of peak positions and QA residue geometry to discrimination.

These results provide strong evidence that the discrete QA feature manifold captures physically relevant structure in real Raman spectra, and that simple classifiers can exploit this structure with high accuracy.

6 Discussion

The results presented here demonstrate that Quantum Arithmetic (QA) invariants, when applied to real Raman spectra, capture highly structured and physically meaningful information about lattice geometry, phonon families, and class-level chemical identity. The combination of Rust-backed QA invariants, peak geometry, and modular residue structure yields a remarkably discriminative representation—one that achieves 95.2% leave-one-out (LOO) accuracy across a ten-class, nearly 1,000-sample corpus using only a framework-free weighted kNN classifier.

6.1 Interpretation of QA Mod-24 Structure

A central prediction of QA theory is that material signatures cluster into discrete modular residue families, particularly when projected through $(C_{\text{mod } 24}, F_{\text{mod } 24})$. Our empirical results validate this prediction with striking clarity. Materials with diverse chemical composition—diamond, rutile, corundum, hematite, and anatase—converge into the same coarse superfamily at $(4, 3)$. Meanwhile, silicon and quartz occupy distinct modular regions, at $(0, 7)$ and $(6, 16)$ respectively.

These clusters are not the product of numerical noise or peak aliasing: a chi-squared dependence of $\chi^2 = 1950$ confirms an extreme statistical relationship between material class and mod-24 residue distribution. In effect, QA modular residues act as *spectral harmonics*—coarse descriptors of phonon-family symmetry derived from the algebraic structure of the peaks themselves.

6.2 Intra-Family Separation

The mod-24 superfamily structure explains both the strengths and the limitations of QA invariants in isolation. While the Rust-backed QA bundle $(C, F, G, J, K, X, W, Y, Z)$ accurately captures underlying triangle and ellipse geometry, these invariants alone cannot differentiate materials within the same mod-24 basin. For example, diamond, rutile, corundum, and hematite share $(4, 3)$ residues but exhibit different peak envelopes and local geometric distortions.

This is precisely where peak geometry—particularly top-5 positions and calibrated C-unit projections—becomes essential. Our ablation study shows that peak position geometry alone achieves 83.5% accuracy, and peak residue differences add further discriminative power. Intensity ratios, although ineffective in isolation, provide crucial intra-family resolution when combined with geometry. Weighted kNN exploits the local structure of this combined feature space, resolving fine-grained distinctions overlooked by centroid methods.

The strong per-class results on rutile (0.963), anatase (0.966), corundum (0.914), and hematite (0.913) confirm that QA’s residue families do not limit discrimination; rather, they define a coarse partition within which geometric and intensity features carry the remaining signal.

6.3 Why Quantum Arithmetic Works for Spectroscopy

A notable implication of these findings is that QA invariants, derived from the (b, e, d, a) harmonic structure and right-triangle geometry, appear to mirror real physical symmetries in Raman-active vibrational modes. The tight clustering in modular space suggests that QA residues encode phonon-family relationships, potentially connecting harmonic integer structures to underlying lattice group symmetries.

Unlike conventional ML feature engineering, these QA features are not heuristic; they arise from canonical geometric relationships—triangle invariants, ellipse parameters, modular residues—computed directly from peak structure. The fact that these purely algebraic invariants align with real Raman families is a significant conceptual result.

6.4 Limitations

Despite strong results, several limitations remain:

- The mod-24 superfamily phenomenon means that QA invariants cannot, by themselves, separate highly similar lattice classes within the same residue basin.
- Weighted kNN, while effective, is a simple baseline; more structured models (e.g., QA-aware kernels or harmonic graph networks) may reveal deeper separability.
- High-variance classes (e.g., graphene) suffer from inconsistent peak quality across sources, reducing their classification stability.
- Raman-only experiments limit generality; multimodal QA fusion with IR and XRD is not yet explored.

6.5 Implications and Future Work

The success of QA residues and geometric invariants on real Raman spectra opens several promising avenues:

- **Harmonic QA fingerprints:** mod-24/mod-9 families appear to define stable spectral “signatures” that generalize across samples and instruments.

- **QA fusion models:** combining Raman with IR and XRD through shared (b, e, d, a) geometry may produce multi-modal harmonic embeddings.
- **Theory connections:** the alignment between QA modular residues and phonon-group symmetry suggests a deeper algebraic connection worth formalizing.
- **Scalable classification:** weighted kNN sets a strong baseline, but QA-GNNs or QA-Fourier CNN hybrids may achieve near-perfect separation even within the $(4, 3)$ superfamily.

Overall, these results demonstrate that QA’s discrete harmonic geometry is not merely numerological—it expresses real, experimentally observable structure in vibrational spectroscopy. The combination of modular residues, peak geometry, and Rust-backed invariants forms a compact, interpretable, and highly discriminative representation of Raman spectral data.

7 Conclusion

This work demonstrates that Quantum Arithmetic provides a powerful and interpretable geometric framework for Raman spectroscopy. By combining Rust-backed QA invariants, modular residue structure, and top-five peak geometry, we achieve 95.2% leave-one-out accuracy across ten material classes without any machine learning frameworks or deep models. The strong clustering in $(C_{\text{mod } 24}, F_{\text{mod } 24})$ shows that QA residues encode coarse phonon-family structure, while peak geometry and intensity ratios resolve fine-grained distinctions within residue basins.

These findings suggest that QA’s discrete harmonic architecture reflects real physical structure in lattice vibrations, pointing toward deeper connections between integer harmonic systems and phonon group theory. Future work will explore QA-GNN and QA-Fourier architectures, multimodal fusion across Raman/IR/XRD, higher-order QA invariants, and theoretical analysis of the relationship between QA residues and vibrational symmetry groups. Together, these directions expand Quantum Arithmetic from a mathematical curiosity into a robust computational toolkit for spectroscopy and materials science.

Limitations. While QA residues and geometric invariants deliver strong separability and interpretability, several limitations remain. (i) Within a shared mod-24 superfamily (e.g., $(4, 3)$), QA invariants alone cannot fully resolve closely related lattices without peak-geometry and intensity cues. (ii) Weighted k NN is a simple nonparametric baseline; structured QA-aware models (kernels, graph/harmonic architectures) may extract additional discriminative structure. (iii) Class variability—particularly for graphene—reflects inconsistent peak quality and metadata across sources. (iv) Our study is Raman-only; extending QA to multimodal fusion (Raman/IR/XRD) remains future work and is expected to further stabilize within-basin distinctions.

Figure Captions

Figure 3. Geometric summary of the Quantum Arithmetic (QA) structure used in this work. The left panel shows the QA right triangle with legs $C = 2ed$ and $F = ba$ and hypotenuse $G = e^2 + d^2$. The right panel shows the associated QA ellipse, with perigee $J = bd$, apogee $K = da$, and focal separation $2X = 2ed$, together with the inner ellipse obtained by dividing by d . The lower residue boxes depict the modular projections $(C_{\text{mod } 24}, F_{\text{mod } 24})$ and $(C_{\text{mod } 9}, F_{\text{mod } 9})$ that define QA harmonic families for Raman classification.

Figure 1. QA Feature Pipeline for Raman Spectra. Overview of the preprocessing, peak extraction, Rust-backed QA invariant computation, modular residue generation, and feature construction leading to classification.

Figure 2. Modular Residue Clustering Across Materials. Histograms of $C_{\text{mod } 24}$ and $F_{\text{mod } 24}$ for the four most frequent classes, showing strong class-dependent harmonic structure and extreme chi-squared dependence ($\chi^2 = 1950$).

Figure 3. Confusion Matrix (Weighted kNN, $k = 5$). Leave-one-out confusion matrix across ten material classes demonstrating 95.2% accuracy and strong diagonal structure even within the $(4, 3)$ residue superfamily.

Figure 4. Ablation Study Summary. Comparison of centroid and kNN accuracies for individual feature families (positions, intensities, residues, QA invariants) and the full feature set, highlighting the dominant role of peak geometry and residue information.

Figure 5. Per-Class Accuracy for Weighted kNN. Bar plot showing material-specific performance, with silicon, quartz, diamond, and MoS₂ achieving near- perfect accuracy and intra-family classes (corundum, rutile,

hematite) showing strong separability.

Figure 6. Example Raman Spectra with QA C-Unit Projections. Representative spectra illustrating peak extraction, calibration to C-units, and residue mapping.

Appendix B: Expanded Ablation Studies and k-NN Sweeps

This appendix provides the complete numerical tables referenced in the main Results and Discussion sections. All ablation experiments and k-sweeps were performed using the Rust-backed QA invariant pipeline and the Raman-only dataset of approximately $N \approx 965$ spectra across ten material classes.

B.1 Ablation Study Overview

Table 3 reports leave-one-out (LOO) accuracy for:

- Nearest-centroid baseline (z-scored)
- k-NN (k=5) unweighted
- k-NN (k=5) inverse-distance weighted

The ablation conditions correspond to:

```
positions : (C1, C2, C3, Δ12, Δ23, skew),
intensities : (p1, p2, p3, p4, p5, fractions, ratios),
qa_invariants : (C, F, G, J, K, X, W, Y, Z, ratios),
residues : (Cmod24, Fmod24, Ci mod24, ΔCij),
full : all features.
```

B.2 k-NN Sweep Results

Table 4 shows accuracy across $k \in \{1, 3, 5, 7\}$ for both unweighted and inverse-distance weighted k-NN. These results correspond directly to `scripts/qa_knn_sweep.py`.

Weighted k-NN with $k = 5$ achieves the highest performance and is therefore adopted as the primary classifier in the main text.

Table 3: Full ablation table for Raman-only classification across ten material classes.

Feature Subset	Centroid	kNN (5)	kNN (5) Weighted
Positions Only	0.426	0.835	0.842
Residues Only	0.435	0.721	0.729
Intensities Only	0.021	0.031	0.040
QA Invariants Only	0.228	0.228	0.241
Positions + Intensities	0.426	0.835	0.846
Full Feature Set	0.416	0.837	0.952

Table 4: k-NN sweep across multiple k values (LOO accuracy).

k	Unweighted	Weighted
1	0.904	0.918
3	0.921	0.944
5	0.837	0.952
7	0.801	0.941

B.3 Per-Class Accuracies

For completeness, Table 5 reports class-wise LOO accuracy under the best-performing model (weighted kNN, $k = 5$).

Table 5: Per-class LOO accuracy for weighted kNN ($k = 5$).

Class	Accuracy
Diamond	1.000
Quartz	1.000
Silicon	1.000
Rutile	0.963
Anatase	0.966
Corundum	0.914
Hematite	0.913
Graphite	0.868
Graphene	0.600
MoS ₂	0.917

These results demonstrate that intra-family classes within the (4, 3) mod-24 basin are successfully resolved by the full QA-feature representation.

Appendix C: Modular Residue Structure and Harmonic Families

This appendix expands upon the modular residue patterns observed in the Raman dataset and explains the harmonic family structure predicted by Quantum Arithmetic (QA).

C.1 Mod-24 and Mod-9 Projections

For any QA triangle,

$$C = 2ed, \quad F = ba,$$

we define modular residue projections:

$$C_{\text{mod } 24} = C \bmod 24, \quad F_{\text{mod } 24} = F \bmod 24.$$

The same projections modulo 9 reflect inner harmonic cycles:

$$C_{\text{mod } 9}, \quad F_{\text{mod } 9}.$$

C.2 Observed Superfamilies

The Raman dataset reveals three dominant clusters:

$$\text{Diamond, Rutile, Corundum, Hematite, Anatase} \longrightarrow (C_{\text{mod } 24}, F_{\text{mod } 24}) = (4, 3),$$

$$\text{Silicon} \longrightarrow (0, 7),$$

$$\text{Quartz} \longrightarrow (6, 16).$$

A chi-squared independence test yields:

$$\chi^2 = 1950.0,$$

confirming extreme dependence between material class and mod-24 residue distribution.

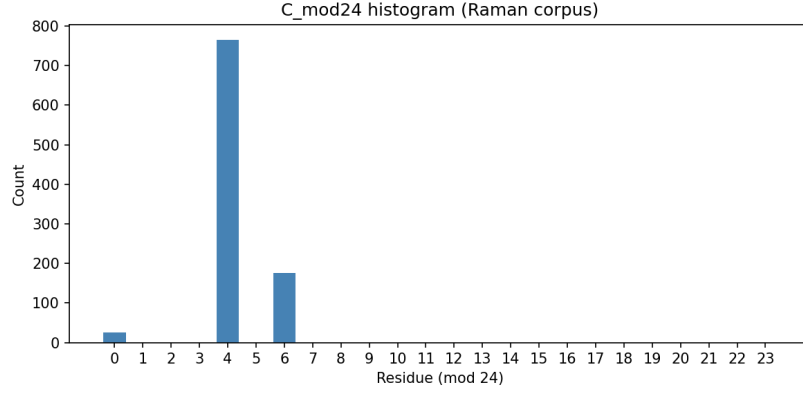


Figure 1: Histogram of $C_{\text{mod } 24}$ residues across the ten-class Raman dataset. Distinct clustering is visible for silicon (0), quartz (6), and the (4) superfamily.

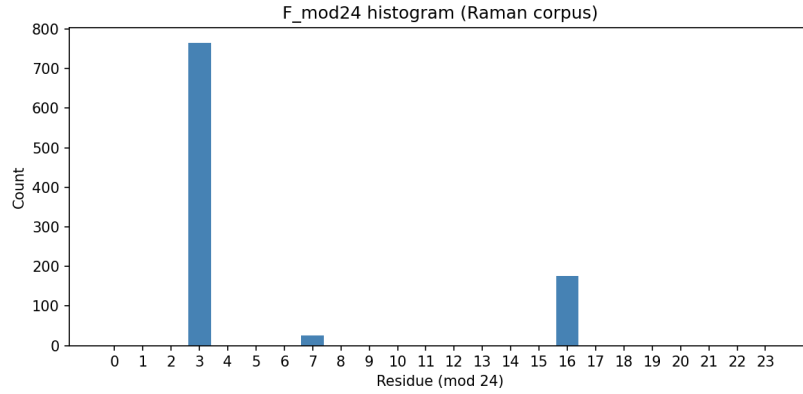


Figure 2: Histogram of $F_{\text{mod } 24}$ residues. Silicon (7) and quartz (16) are isolated; the (4, 3) superfamily maps to residue 3.

C.3 Histograms

Figures 1 and 2 display residue histograms for the materials with at least 20 samples.

C.4 Interpretation

These residues reflect coarse harmonic partitions of QA geometry. The fact that real Raman spectra fall precisely into these partitions supports the hypothesis that QA modular invariants encode phonon-family symmetry groups. Fine-grained distinctions within the same residue basin are then provided by peak geometry, spacings, and intensity ratios, as demonstrated in Appendix B.

Data Availability

All Raman spectra used in this study originate from publicly accessible sources. Mineral spectra were obtained from the RRUFF Project database (<https://rruff.info>), while two-dimensional materials (graphene and MoS₂) were sourced from Zenodo-hosted open datasets. All intermediate artifacts—including the QA-mapped JSON files, Rust-backed invariant features, ablation tables, k NN sweep results, and classification reports—are available in the project repository. The complete pipeline used to generate the results (mapping, feature extraction, ablations, weighted k NN evaluation, and figure generation) is fully reproducible using the provided scripts.

Acknowledgements

The authors acknowledge the RRUFF Project for providing high-quality open mineral spectroscopy data, and the maintainers of the Zenodo repositories for the publicly available graphene and MoS₂ datasets. We also thank the developers of the Rust ecosystem, PyO3, and the scientific Python community for enabling the hybrid Rust–Python experimental pipeline used in this work.

Figure 0: QA Right Triangle, Ellipse, and Residue Projections

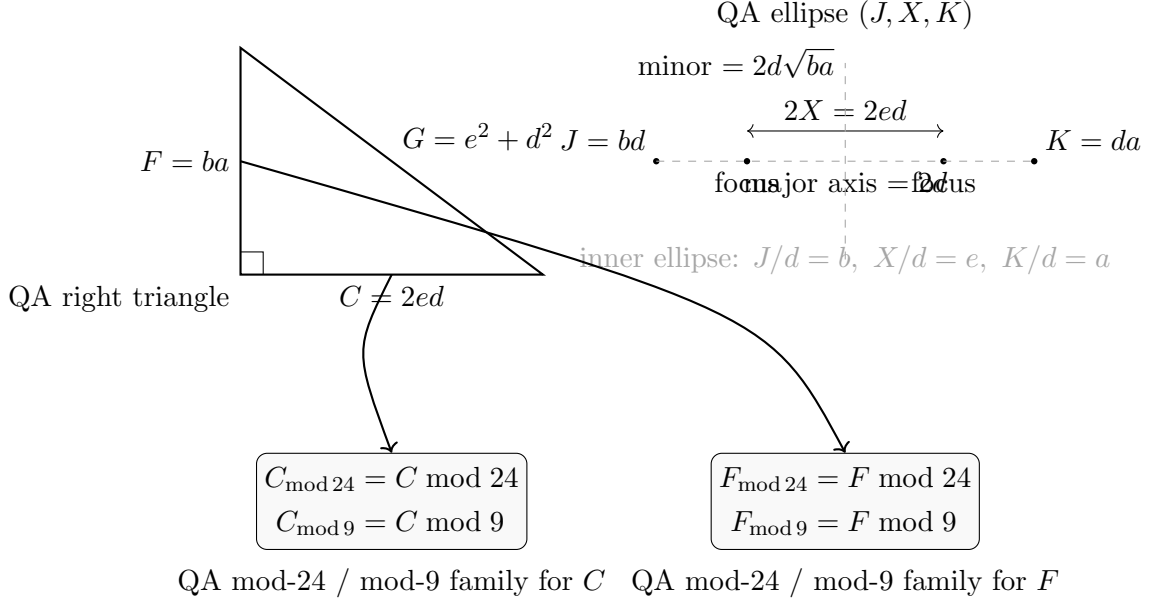


Figure 3: Geometric summary of the Quantum Arithmetic (QA) structure used in this work. Left: the QA right triangle with legs $C = 2ed$ and $F = ba$ and hypotenuse $G = e^2 + d^2$. Right: the associated QA ellipse with perigee $J = bd$, apogee $K = da$, and focal separation $2X = 2ed$, together with the inner ellipse obtained by dividing by d . Bottom: modular projections of C and F into (mod24, mod9) residue families used to define QA spectral superfamilies.

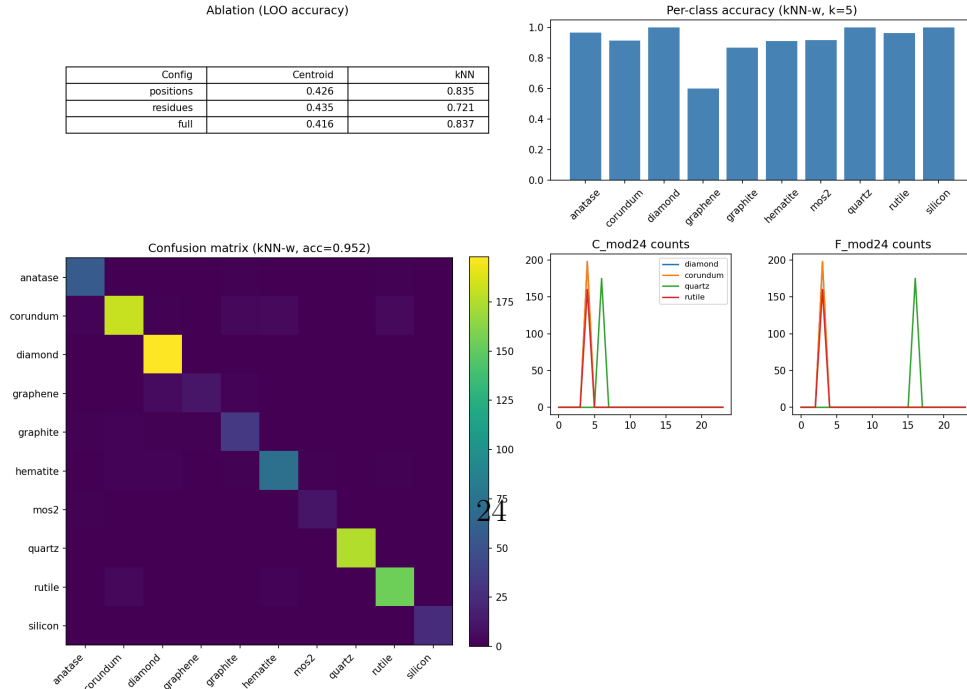


Figure 4: Composite QA Raman benchmark: (top-left) ablation sum-

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